

NATHAN INBAR

Victoria, British Columbia

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SUMMARY

Aspiring research-focused computer scientist with a background in knowledge graph systems, GraphRAG architectures, and distributed ingestion pipelines. Experience building provenance-preserving graph backends and evaluating structured retrieval methods. Research interests include semantic clustering, incremental knowledge construction, and graph-based retrieval over domain-specific knowledge.

EDUCATION

Bachelor of Science in Computer Science (BSc)
University of Victoria, Victoria, BC, CA

Expected Aug 2026

TECHNICAL SKILLS

AI / ML / NLP: Knowledge Graphs, DSPy, LiteLLM, PyTorch

Databases & Graphs: Postgres, Memgraph / Neo4j, Redis

Infrastructure & Cloud: Docker, AWS (S3, Bedrock, API Gateway, IAM)

Languages: Python, SQL, Cypher, Bash, C#

PUBLICATIONS

Tech Tales: Digital Storytelling with the Royal BC Museum

SIGGRAPH Appy Hour '25, ACM SIGGRAPH Conference Proceedings

DOI: 10.1145/3721260.3733981

EXPERIENCE

Junior Software Development Intern (Knowledge Graph Systems)
Pacific Coastal Computing, Victoria, BC

June - Aug 2025

Key Contributions

- Led architectural migration from a monolithic Python backend to a set of independently deployable microservices for a GraphRAG knowledge graph system
- Implemented message-based ingestion pipeline using Redis Streams, enabling backpressure-aware horizontal scaling across containers
- Built a Memgraph-backed graph layer using complex OpenCypher queries to support idempotent node upserts and project-scoped graph filtering
- Implemented LLM-based triple extraction using DSPy with asynchronous execution and rate-limiting
- Invited to discuss system architecture and ownership on a developer podcast
URL: <https://youtu.be/1m3-Q9f6VT0> (Timestamp 2:30)

ACADEMIC PROJECTS

GraphRAG Methodology Evaluation (Ongoing Research)

- Designing comparative evaluation of structured knowledge graph generation and retrieval methods, including LeanRAG, HyperGraph, and VectorRAG
- Evaluating differences in semantic clustering quality differences under exposure to unseen and evolving datasets
- Targeting academic conference submission (Canadian AI 2026)